Spring Semester 2025–2026

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 5 – SOLUTIONS

1. (a) For the general ARMA process, $\phi(B)X_t = \theta(B)Z_t$, deduce that the spectral density function is given by

$$g(w) = \frac{\sigma_z^2}{2\pi}\frac{\theta(e^{-iw})\theta(e^{iw})}{\phi(e^{-iw})\phi(e^{iw})}, \quad -\pi < w \leq \pi.$$

SOLUTION:

The $ARMA(p, q)$ process can be written

$$\begin{aligned} X_t &= \frac{\theta(B)}{\phi(B)}Z_t \\ &= \psi(B)Z_t \end{aligned}$$

where $\psi(B) = \theta(B)/\phi(B)$.

We know that the spectral density function of $\{X_t\}$ is

$$\begin{aligned} g(w) &= \frac{\sigma_z^2}{2\pi}\psi(e^{iw})\psi(e^{-iw}) \\ &= \frac{\sigma_z^2}{2\pi}|\psi(e^{iw})|^2 \\ &= \frac{\sigma_z^2}{2\pi}\left|\frac{\theta(e^{iw})}{\phi(e^{iw})}\right|^2 \\ &= \frac{\sigma_z^2}{2\pi}\left(\frac{|\theta(e^{iw})|}{|\phi(e^{iw})|}\right)^2 \quad \text{(using properties of complex numbers)} \\ &= \frac{\sigma_z^2}{2\pi}\frac{|\theta(e^{iw})|^2}{|\phi(e^{iw})|^2} \\ &= \frac{\sigma_z^2}{2\pi}\frac{\theta(e^{iw})\theta(e^{-iw})}{\phi(e^{iw})\phi(e^{-iw})}. \end{aligned}$$

(b) Hence verify the form of $g(w)$ given in lectures for a stationary AR(2) process.

SOLUTION:

For an AR(2) model, $(1 - \phi_1 B - \phi_2 B^2)X_t = Z_t$, $\theta(z) \equiv 1$ and so

$$g(w) = \frac{\sigma_z^2}{2\pi}\frac{1}{(1 - \phi_1 e^{-iw} - \phi_2 e^{-2iw})(1 - \phi_1 e^{iw} - \phi_2 e^{2iw})}.$$

Then

$$\begin{aligned}
&(1-\phi_1 e^{-iw}-\phi_2 e^{-2iw})(1-\phi_1 e^{iw}-\phi_2 e^{2iw}) \\
&\quad = 1+\phi_1^2+\phi_2^2-\phi_1(e^{-iw}+e^{iw})-\phi_2(e^{-2iw}+e^{2iw})+\phi_1\phi_2(e^{-iw}+e^{iw}) \\
&\quad = 1+\phi_1^2+\phi_2^2-2\phi_1\cos w-2\phi_2\cos 2w+2\phi_1\phi_2\cos w.
\end{aligned}$$

Therefore in the AR(2) case,

$$g(w)=\frac{\sigma_z^2}{2\pi}\frac{1}{1+\phi_1^2+\phi_2^2-2\phi_1(1-\phi_2)\cos w-2\phi_2\cos 2w}.$$

(c) Show also that for a (weakly) stationary ARMA$(1,1)$ process

$$(1-\phi B)X_t=(1+\theta B)Z_t$$

we have

$$g(w)=\frac{\sigma_z^2}{2\pi}\frac{\{1+\theta^2+2\theta\cos(w)\}}{\{1+\phi^2-2\phi\cos(w)\}}, \qquad -\pi<w\le\pi.$$

SOLUTION:

For a stationary ARMA$(1,1)$ process, $\theta(z)=1+\theta z$ and $\phi(z)=1-\phi z$, so

$$\theta(e^{-iw})\theta(e^{iw})=(1+\theta e^{-iw})(1+\theta e^{iw})=1+\theta^2+2\theta\cos w,$$

and, using similar calculations,

$$\phi(e^{-iw})\phi(e^{iw})=1+\phi^2-2\phi\cos w.$$

Consequently

$$g(w)=\frac{\sigma_z^2}{2\pi}\frac{(1+\theta^2+2\theta\cos w)}{(1+\phi^2-2\phi\cos w)}.$$

2. A linear filter, applied to the stationary process $\{Y_t\}$, is defined

$$X_t=Y_t-Y_{t-4}$$

(a) Determine the transfer function for this filter and sketch the squared gain.

SOLUTION:

The transfer function is given by

$$h(w)=\sum_{j=-\infty}^{\infty}b_j e^{-iwj}.$$

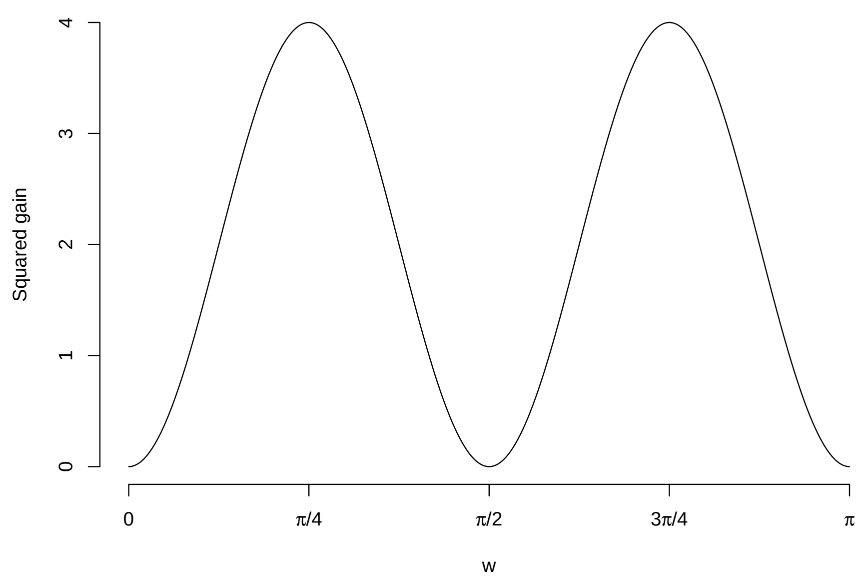
Here $b_0 = 1, b_4 = -1$ and $b_j = 0$ for all other values of j. Then

$$h(w) = 1 - e^{-i4w}.$$

The squared gain is given by

$$\begin{aligned} |h(w)|^2 &= h(w)\overline{h(w)} \\ &= (1 - e^{-i4w})(1 - e^{i4w}) \\ &= 2 - (e^{i4w} + e^{-i4w}) \\ &= 2(1 - \cos 4w). \end{aligned}$$

A plot of the squared gain is shown below



(b) Hence discuss the effect of the filter.

SOLUTION:

For $w \in [0, \pi]$, the squared gain is zero at $w = 0$, $w = \pi/2$ and $w = \pi$. The squared gain attains a maximum of 4 when $w = \pi/4$ and $w = 3\pi/4$. Therefore, frequencies in neighbourhoods of $w = 0$, $w = \pi/2$ and $w = \pi$ are damped down, while frequencies in the neighbourhoods of $w = \pi/4$ and $w = 3\pi/4$ are amplified.

3. Suppose a linear filter is defined by

$$X_t = Y_t - 2Y_{t-1}\cos(\bar{\omega}) + Y_{t-2},$$

where $\bar{\omega}$ is a fixed frequency.

(a) Determine the form of the transfer function for this filter.

SOLUTION:

In this example, $b_0 = 1$, $b_1 = -2\cos\bar{\omega}$ and $b_2 = 1$. Consequently

$$\begin{aligned} h(w) &= \sum_u b_u e^{-iuw} \\ &= 1 - 2\cos\bar{\omega} e^{-iw} + e^{-i2w} \\ &= 1 - 2\cos\bar{\omega}\cos w + \cos 2w + i(2\cos\bar{\omega}\sin w - \sin 2w). \end{aligned}$$

(b) For which frequencies ω does the spectral density function of $\{X_t\}$ equal zero?

SOLUTION:

The squared gain is given by

$$\begin{aligned} |h(w)|^2 &= (1 - 2\cos\bar{\omega} e^{-iw} + e^{-i2w})(1 - 2\cos\bar{\omega} e^{iw} + e^{i2w}) \\ &= 1 + 4\cos^2\bar{\omega} + 1 + e^{-i2w} + e^{i2w} - 2(e^{-iw} + e^{iw})\cos\bar{\omega} - 2(e^{-iw} + e^{iw})\cos\bar{\omega} \\ &= 2(1 + 2\cos^2\bar{\omega} + 2\cos^2 w - 1 - 4\cos\bar{\omega}\cos w) \\ &= 4(\cos w - \cos\bar{\omega})^2. \end{aligned}$$

So, restricting attention to $w \in [0, \pi]$, the spectral density function of $\{X_t\}$ is zero when $w = \bar{\omega}$.

4. It is known that the spectral density function of a certain MA(2) process $\{X_t\}$ is given by

$$g(\omega) = \frac{1}{2\pi}\left\{19 + 5\cos(\omega) - 6\cos(2\omega)\right\}.$$

If $X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$, where $\{Z_t\}$ is a white noise process with variance σ_z^2, show that

$$1 + \theta_1^2 + \theta_2^2 = \frac{19}{\sigma_z^2}, \qquad 2\theta_1(1 + \theta_2) = \frac{5}{\sigma_z^2} \qquad \text{and} \qquad \theta_2 = -\frac{3}{\sigma_z^2}.$$

SOLUTION:

We know that the MA(2) process is written

$$\begin{aligned} X_t &= Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} \\ &= (1 + \theta_1 B + \theta_2 B^2) Z_t \\ &= \theta(B) Z_t \end{aligned}$$

and the spectral density function is given by

$$\begin{aligned}
g(\omega) &= \frac{\sigma_z^2}{2\pi}\theta(e^{-i\omega})\theta(e^{i\omega}) \\
&= \frac{\sigma_z^2}{2\pi}(1+\theta_1 e^{-i\omega}+\theta_2 e^{-i2\omega})(1+\theta_1 e^{i\omega}+\theta_2 e^{i2\omega}) \\
&= \frac{\sigma_z^2}{2\pi}\{1+\theta_1^2+\theta_2^2+\theta_1(e^{-i\omega}+e^{i\omega})+\theta_2(e^{-i2\omega}+e^{i2\omega})+\theta_1\theta_2(e^{-i\omega}+e^{i\omega})\} \\
&= \frac{\sigma_z^2}{2\pi}(1+\theta_1^2+\theta_2^2+2\theta_1(1+\theta_2)\cos\omega+2\theta_2\cos 2\omega).
\end{aligned}$$

Equating coefficients of $\cos k\omega$ with $k = 0, 1, 2$ we obtain

$$19 = \sigma_z^2(1+\theta_1^2+\theta_2^2), \qquad 5 = 2\sigma_z^2\theta_1(1+\theta_2) \quad \text{and} \quad -6 = 2\sigma_z^2\theta_2,$$

which are equivalent to the three identities in the question.

5. The process $\{X_t\}$ is an $ARMA(1,2)$ process

$$X_t = \phi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 and $|\phi| < 1$, $|\theta_1| < 1$ and $|\theta_2| < 1$. Determine the spectral density function of this process.

SOLUTION:

We write the process as

$$\phi(B)X_t = \theta(B)Z_t$$

where

$$\begin{aligned}
\phi(B) &= 1 - \phi_1 B \\
\theta(B) &= 1 + \theta_1 B + \theta_2 B^2.
\end{aligned}$$

Using the result from question 1, we know that

$$g(w) = \frac{\sigma_z^2}{2\pi}\frac{\theta(e^{iw})\theta(e^{-iw})}{\phi(e^{iw})\phi(e^{-iw})}.$$

Then

$$\begin{aligned}
g(w) &= \frac{\sigma_z^2}{2\pi}\left[\frac{(1+\theta_1 e^{iw}+\theta_2 e^{i2w})(1+\theta_1 e^{-iw}+\theta_2 e^{-i2w})}{(1-\phi_1 e^{iw})(1-\phi_1 e^{-iw})}\right] \\
&= \frac{\sigma_z^2}{2\pi}\left[\frac{1+\theta_1^2+\theta_2^2+2\theta_1(1+\theta_2)\cos w+2\theta_2\cos 2w}{1+\phi_1^2-2\phi_1\cos w}\right]
\end{aligned}$$

6. Suppose that $\{X_t\}$ and $\{Y_t\}$ are independent stationary processes with mean zero and autocovariance functions $\gamma_X(h)$ and $\gamma_Y(h)$, respectively. In addition, the spectral density

functions of $\{X_t\}$ and $\{Y_t\}$ are $g_X(w)$ and $g_Y(w)$, respectively.

The process $\{V_t\}$ is such that

$$V_t = X_t + Y_t.$$

(a) Show that the process $\{V_t\}$ is also stationary and has spectral density function $g_V(w)$ such that

$$g_V(w) = g_X(w) + g_Y(w).$$

SOLUTION:

Consider $\gamma_V(h)$ where

$$\gamma_V(h) = \text{Cov}(V_t, V_{t+h})$$

Since $\mathbb{E}(V_t) = 0$, it follows that

$$\begin{aligned}
\mathbb{E}(V_t V_{t+h}) = \gamma_V(h) &= \mathbb{E}\left((X_t + Y_t)(X_{t+h} + Y_{t+h})\right) \\
&= \mathbb{E}\left(X_t X_{t+h} + Y_t Y_{t+h} + X_{t+h} Y_t + X_t Y_{t+h}\right) \\
&= \mathbb{E}\left(X_t X_{t+h}\right) + \mathbb{E}\left(Y_t Y_{t+h}\right) \\
&= \gamma_X(h) + \gamma_Y(h)
\end{aligned}$$

and $\{V_t\}$ is stationary since neither $\gamma_X(h)$ nor $\gamma_Y(h)$ depends on h.

Now, the spectral density function of $\{V_t\}$ is

$$\begin{aligned}
g_V(w) &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma_V(h) e^{iwh} \\
&= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} (\gamma_X(h) + \gamma_Y(h)) e^{iwh} \\
&= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma_X(h) e^{iwh} + \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma_Y(h) e^{iwh} \\
&= g_X(w) + g_Y(w), \quad \text{as required.}
\end{aligned}$$

(b) Suppose that $\{X_t\}$ is an $AR(1)$ process, defined

$$X_t = \phi X_{t-1} + W_t$$

where $|\phi| < 1$ and $\{W_t\}$ is a white noise process with variance σ^2. In addition, $\{Y_t\}$ is a white noise process with variance σ^2. Determine the spectral density function of

$$V_t = X_t + Y_t.$$

SOLUTION:

We use the result from part (a), denoting $g_X(w)$ and $g_Y(w)$ to be the spectral density functions of $\{X_t\}$ and $\{Y_t\}$ respectively.

Since

$$\begin{aligned}
(1-\phi B)X_t &= W_t \\
\implies \phi(B)X_t &= W_t \quad (\text{where } \phi(B) = 1-\phi B) \\
X_t &= \frac{1}{\phi(B)}W_t
\end{aligned}$$

Then

$$\begin{aligned}
g_X(w) &= \frac{\sigma^2}{2\pi}\frac{1}{\phi(e^{iw})}\frac{1}{\phi(e^{-iw})} \\
&= \frac{\sigma^2}{2\pi(1-\phi e^{iw})(1-\phi e^{-iw})} \\
&= \frac{\sigma^2}{2\pi(1+\phi^2-2\phi\cos w)}.
\end{aligned}$$

Also, since $\{Y_t\}$ is a white noise process

$$g_Y(w) = \frac{\sigma^2}{2\pi}.$$

Then, using the result from (a)

$$\begin{aligned}
g_V(w) &= g_X(w) + g_Y(w) \\
&= \frac{\sigma^2}{2\pi}\left(1 + \frac{1}{1+\phi^2-2\phi\cos w}\right) \\
&= \frac{\sigma^2\left(2+\phi^2-2\phi\cos w\right)}{2\pi\left(1+\phi^2-2\phi\cos w\right)}
\end{aligned}$$

7. (a) Derive the spectral density functions of

   (i) An MA(1) process $X_t = Z_t + \theta Z_{t-1}$.

   SOLUTION:

   Since

$$X_t = \theta(B)Z_t$$

where $\theta(B) = 1 + \theta B$, we know that the spectral density function is

$$\begin{aligned} g_X(w) &= \frac{\sigma_z^2}{2\pi}\theta(e^{iw})\theta(e^{-iw}) \\ &= \frac{\sigma_z^2}{2\pi}(1+\theta e^{iw})(1+\theta e^{-iw}) \\ &= \frac{\sigma_z^2}{2\pi}\left[1+\theta^2+\theta(e^{iw}+e^{-iw})\right] \\ &= \frac{\sigma_z^2}{2\pi}\left[1+\theta^2+2\theta\cos w\right]. \end{aligned}$$

(ii) The process $\{X_t\}$ where

$$(1-\rho B)^2 X_t = Z_t$$

with $|\rho| < 1$.

SOLUTION:

Here

$$\begin{aligned} X_t &= \frac{1}{(1-\rho B)^2} Z_t \\ &= \psi(B) Z_t \end{aligned}$$

where $\psi(B) = (1-\rho B)^{-2}$.

The spectral density function is given by

$$\begin{aligned} g(w) &= \frac{\sigma_z^2}{2\pi}\psi(e^{iw})\psi(e^{-iw}) \\ &= \frac{\sigma_z^2}{2\pi}(1-\rho e^{iw})^{-2}(1-\rho e^{-iw})^{-2} \\ &= \frac{\sigma_z^2}{2\pi}\left[\frac{1}{(1-\rho e^{iw})(1-\rho e^{-iw})}\right]^2 \\ &= \frac{\sigma_z^2}{2\pi(1+\rho^2-2\rho\cos w)^2} \end{aligned}$$

Here $\{Z_t\}$ is a white noise process with variance σ_z^2.

(b) Suppose that $\{X_t\}$ is the linear filter

$$X_t = \frac{2}{3}Y_t + \frac{1}{3}Y_{t-1}$$

and $\{Y_t\}$ is a stationary process with mean μ and spectral density function $g_Y(w) = \sigma_z^2(1-\beta\cos w)$. Derive the mean and spectral density function of $\{X_t\}$.

SOLUTION:

The mean of $\{X_t\}$ is given by

$$\begin{aligned}\mathbb{E}(X_t) &= \frac{2}{3}\mathbb{E}(Y_t) + \frac{1}{3}\mathbb{E}(Y_{t-1}) \\ &= \mu.\end{aligned}$$

The transfer function is

$$h(w) = \sum_{j=-\infty}^{\infty} b_j e^{-iwj}$$

where $b_0 = \frac{2}{3}$, $b_1 = \frac{1}{3}$ and $b_j = 0$ otherwise. Then

$$\begin{aligned}h(w) &= \frac{2}{3} + \frac{1}{3}e^{-iw} \\ \implies |h(w)|^2 &= \frac{1}{9}(2 + e^{-iw})(2 - e^{iw}) \\ &= \frac{1}{9}(5 + 2(e^{iw} + e^{-iw}) \\ &= \frac{1}{9}(5 + 4\cos w)\end{aligned}$$

The spectral density function of $\{X_t\}$ is

$$\begin{aligned}g_X(w) &= |h(w)|^2 g_Y(w) \\ &= |h(w)|^2 \sigma_z^2 (1 - \beta \cos w) \\ &= \frac{\sigma_z^2}{9}(5 + 4\cos w)(1 - \beta\cos w)\end{aligned}$$

8. The process $\{X_t\}$ is the AR(1) process

$$X_t = \frac{1}{2}X_{t-1} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2. Suppose that $\{X_t\}$ is smoothed using the following linear filter (a moving average on three points) where

$$Y_t = \frac{1}{3}\left[X_{t+1} + X_t + X_{t-1}\right].$$

(a) Determine the spectral density function of $\{Y_t\}$.

SOLUTION:

Using the result from 6(b), the spectral density function of $\{X_t\}$ is

$$\begin{aligned}g_X(w) &= \frac{\sigma_z^2}{2\pi\left(1 + \left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)\cos w\right)} \\ &= \frac{\sigma_z^2}{2\pi\left(\frac{5}{4} - \cos w\right)} \\ &= \frac{2\sigma_z^2}{\pi\left(5 - 4\cos w\right)}\end{aligned}$$

The transfer function is given by

$$
\begin{aligned}
h(w) &= \frac{1}{3}\left(1+e^{iw}+e^{-iw}\right) \\
&= \frac{1}{3}\left(1+2\cos w\right)
\end{aligned}
$$

The squared gain is

$$
\begin{aligned}
|h(w)|^2 &= \left[\frac{1}{3}\left(1+2\cos w\right)\right]^2 \\
&= \frac{1}{9}\left(1+4\cos w+4\cos^2 w\right).
\end{aligned}
$$

Hence, the spectral density of $\{Y_t\}$ is

$$
\begin{aligned}
g_Y(w) &= |h(w)|^2 g_X(w) \\
&= \frac{1}{9}\left(1+4\cos w+4\cos^2 w\right)\frac{2\sigma_z^2}{\pi\left(5-4\cos w\right)} \\
&= \frac{2\sigma_z^2}{9\pi}\left[\frac{1+4\cos w+4\cos^2 w}{5-4\cos w}\right].
\end{aligned}
$$

(b) Sketch the spectral density function and squared gain for values of the frequency, w, where $w \in [0, \pi]$.

SOLUTION:

The squared gain may be written

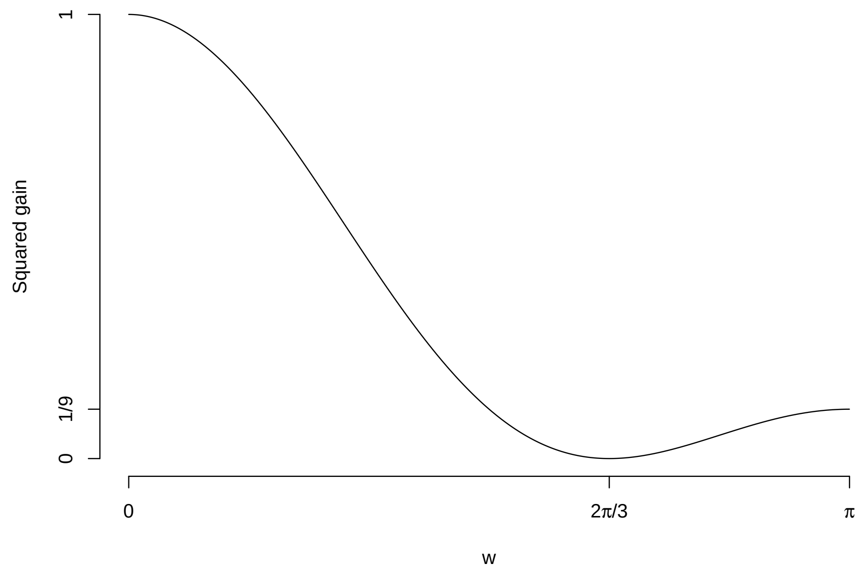
$$
\begin{aligned}
|h(w)|^2 &= \frac{4}{9}\left(\frac{1}{4}+\cos w+\cos^2 w\right) \\
&= \frac{4}{9}\left(\frac{1}{4}+\left(\cos w+\frac{1}{2}\right)^2-\frac{1}{4}\right) \\
&= \frac{4}{9}\left(\cos w+\frac{1}{2}\right)^2
\end{aligned}
$$

Clearly, $|h(w)|^2$ is maximised where $\cos w = +1$ (i.e. $w = 0$) and minimised where $\cos w = -\frac{1}{2}$ (i.e. where $w = \frac{2\pi}{3}$)

A plot of the squared gain against the frequency is shown below.

The spectral density function may be written

$$
\begin{aligned}
g_Y(w) &= \frac{4}{9}\left(\cos w+\frac{1}{2}\right)^2 \times \frac{2\sigma_z^2}{\pi\left(5-4\cos w\right)} \\
&= \frac{8\sigma_z^2}{9\pi}\frac{(\cos w+\frac{1}{2})^2}{5-4\cos w}
\end{aligned}
$$



We can substitute $w = 0$ and $w = \pi$ into the above to deduce that

$$g_Y(0) = \frac{2\sigma_z^2}{\pi};$$
$$g_Y(\pi) = \frac{2\sigma_z^2}{81\pi}.$$

We determine turning points of the function.

Using the quotient rule, let

$$u = \left(\cos w + \frac{1}{2}\right)^2; u' = -2\sin w\left(\cos w + \frac{1}{2}\right);$$
$$v = 5 - 4\cos w; v' = 4\sin w.$$

Then

$$\begin{aligned} g'(w) &= \frac{8\sigma_z^2}{9\pi}\left[\frac{-2\sin w\left(\cos w + \frac{1}{2}\right)(5-4\cos w) - 4\sin w\left(\cos w + \frac{1}{2}\right)^2}{(5-4\cos w)^2}\right] \\ &= \frac{8\sigma_z^2}{9\pi}\left[\frac{-2\sin w\left(\cos w + \frac{1}{2}\right)\left(5 - 4\cos w + 2\left(\cos w + \frac{1}{2}\right)\right)}{(5-4\cos w)^2}\right] \\ &= \frac{8\sigma_z^2}{9\pi}\left[\frac{-\sin w\,(2\cos w + 1)(6 - 2\cos w)}{(5-4\cos w)^2}\right]. \end{aligned}$$

Set

$$\begin{aligned} g'(w) &= 0 \\ \implies \sin w\,(2\cos w + 1)(6 - 2\cos w) &= 0 \\ \implies \sin w = 0 \quad \text{or} \quad \cos w = -\frac{1}{2} \quad &\text{or} \quad \cos w = 3. \end{aligned}$$

Clearly, $\cos w \neq 3$ and, hence, $\sin w = 0$ or $\cos w = -\frac{1}{2}$.

Then $w = 0, \pi$ or $w = \frac{2\pi}{3}$. We know from the squared gain that a local minimum will occur at $w = \frac{2\pi}{3}$. It is also fairly straightforward to deduce that the spectral density is decreasing on $[0, 2\pi/3)$ and increasing on $[2\pi/3, \pi]$. Hence, a sketch of the spectral density function is shown below.